Customer Churn — Exploratory Data Analysis

A Data-Driven Study of Subscriber Attrition for a Telecom Business

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Task:	Task 3 — Customer Churn EDA
Dataset:	Telco Customer Churn (IBM Sample Dataset)
Tool used:	Python (Pandas, Seaborn, Matplotlib) on Google Colab

1. Introduction

Customer churn — when a subscriber stops using a company's service — is one of the most important problems for any subscription-based business to understand, since retaining an existing customer is almost always cheaper than acquiring a new one. This report presents an exploratory data analysis (EDA) of a telecom customer dataset, with the goal of identifying which customers are most likely to churn and what factors are driving that behavior. The analysis covers data cleaning, univariate and bivariate exploration, correlation analysis, and a final set of business recommendations based on the patterns found in the data.

2. Dataset Details

The dataset used is the Telco Customer Churn dataset, originally published as part of IBM's Sample Data Sets and widely used as a benchmark dataset for churn analysis. Each row represents one customer, and the dataset includes demographic information, account details, the services each customer is subscribed to, and a target column indicating whether that customer churned.

Property	Value
Total customers (before cleaning)	7,043
Total columns	21
Target column	Churn (Yes / No)
Feature types	Mix of numerical (tenure, MonthlyCharges, TotalCharges) and categorical (Contract, PaymentMethod, InternetService, etc.)

3. Data Cleaning Summary

Before analysis, the dataset was checked for incorrect data types, missing values, duplicate rows, and outliers. The following cleaning steps were applied:

- **Data type correction:** The TotalCharges column was stored as text instead of a number. It was converted to a numeric type so it could be used in calculations and charts.
- **Missing values:** Converting TotalCharges to numeric revealed 11 rows with missing values (these were blank entries hidden inside the text column). These 11 rows were removed, leaving 7,032 clean customer records.
- **Duplicate rows:** The dataset was checked for duplicate rows; none were found.
- **Irrelevant columns:** The customerID column was removed since it is a unique identifier with no analytical value.
- **Outlier check:** An IQR (Interquartile Range) based outlier check was run on tenure,

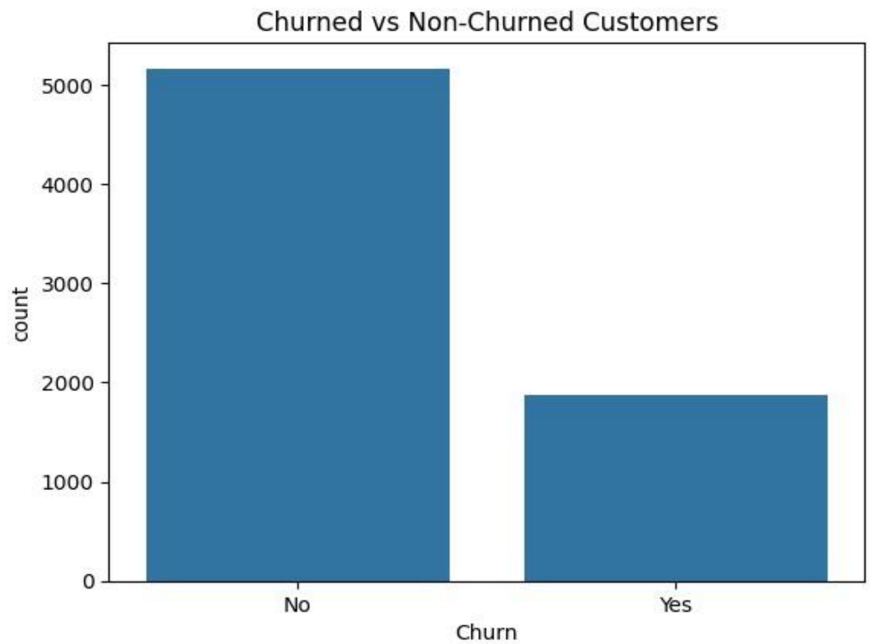
MonthlyCharges, and TotalCharges. No outliers were found in any of the three columns, confirming the numerical data is well-behaved with no extreme or erroneous values.

Final cleaned dataset size: 7,032 rows × 20 columns.

4. Exploratory Data Analysis (EDA) Findings

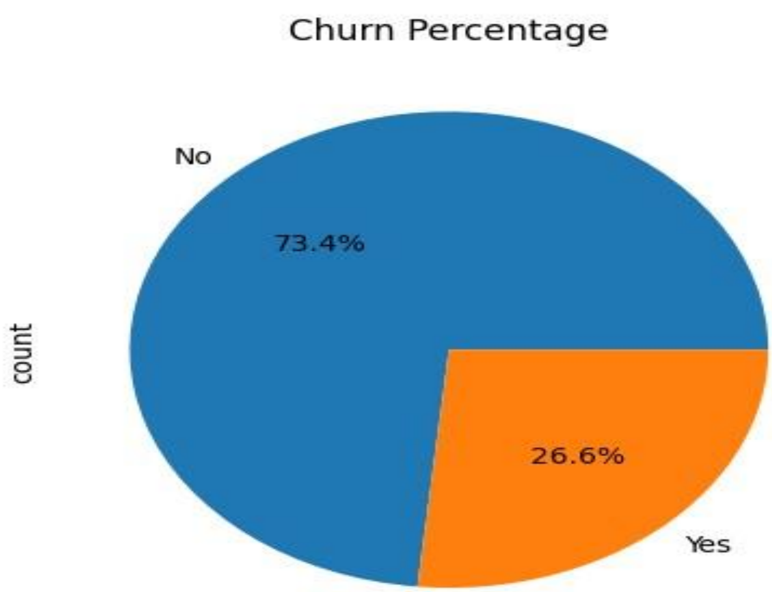
The cleaned dataset was explored using a series of univariate (single-variable) and bivariate (two-variable) visualizations to understand the distribution of key features and how each one relates to churn.

4.1 Churn Distribution (Count)



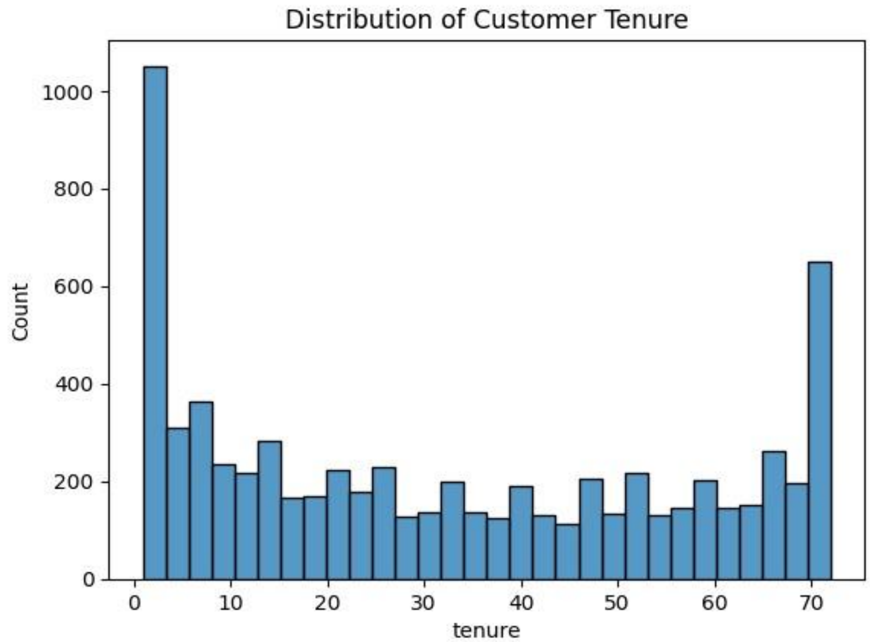
The overall customer base is split between churned and non-churned customers, with non-churned customers forming the clear majority.

4.2 Churn Percentage



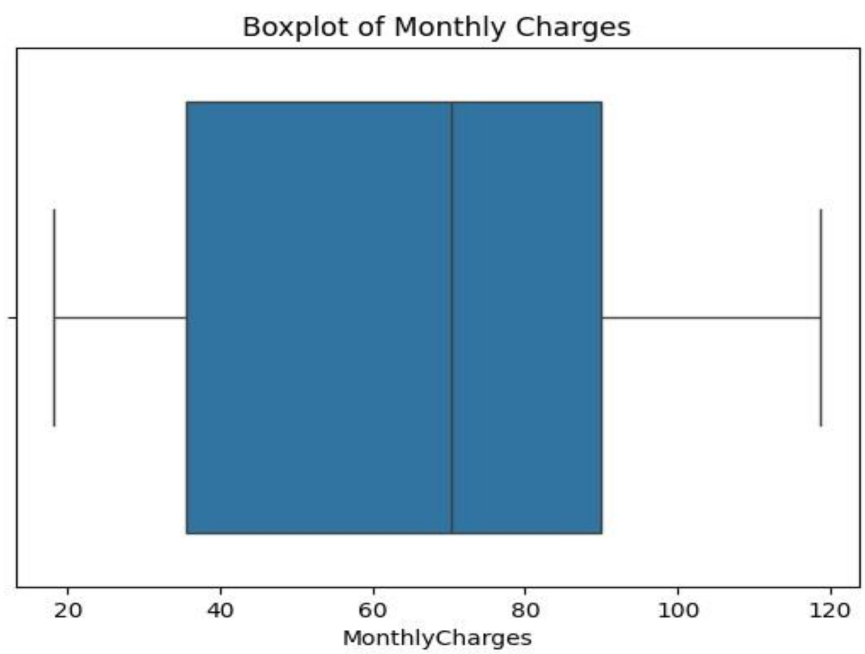
26.6% of customers in the dataset churned, while 73.4% remained active. This confirms that churn, while not the majority outcome, affects a substantial portion of the customer base.

4.3 Distribution of Customer Tenure



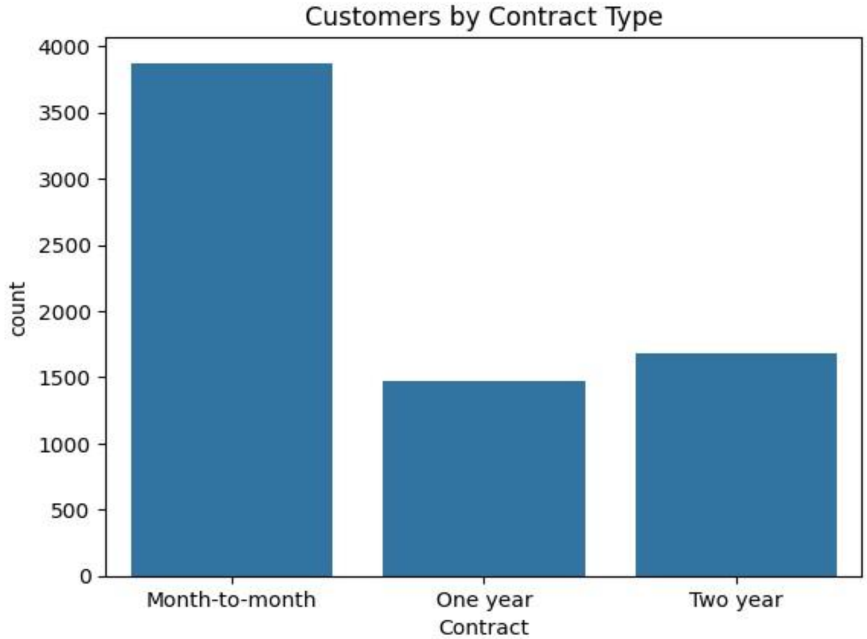
Customer tenure is spread fairly widely, with a noticeable concentration of customers at very low tenure (new customers) and another build-up near the maximum tenure (long-term, loyal customers).

4.4 Boxplot of Monthly Charges



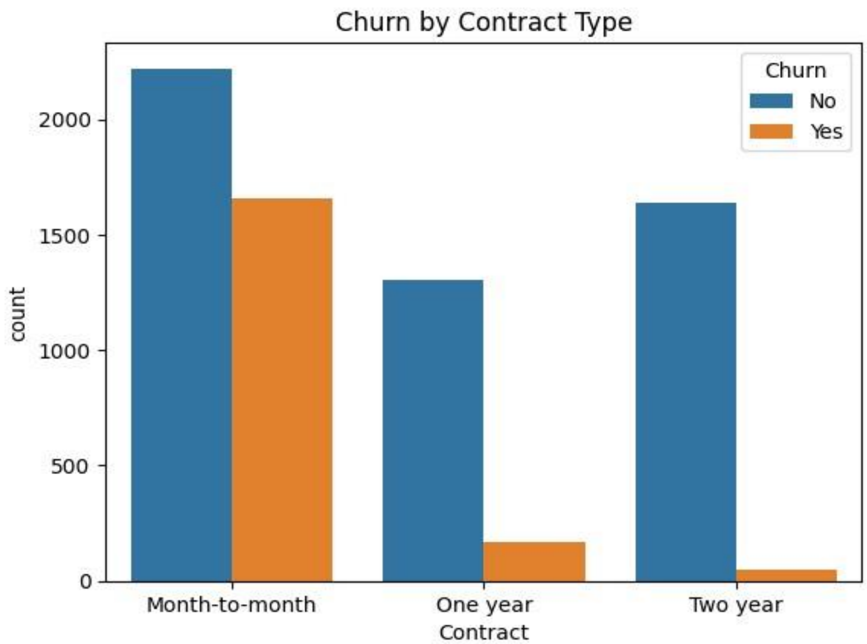
The boxplot shows a reasonably even spread of monthly charges with no points beyond the whiskers, confirming the earlier IQR check that there are no outliers in this column.

4.5 Customers by Contract Type



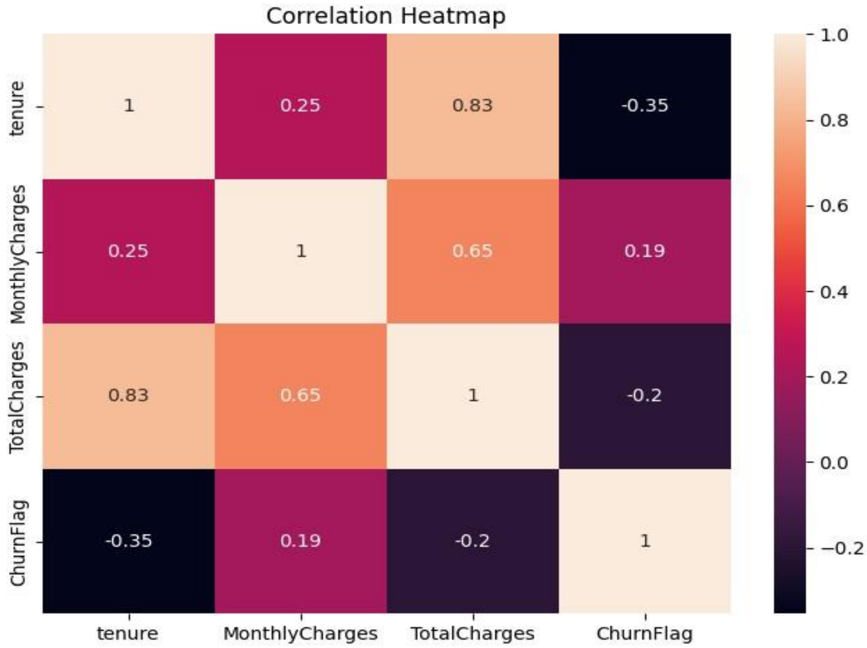
The majority of customers are on month-to-month contracts, followed by two-year and then one-year contracts. This is an important segment to examine against churn.

4.6 Churn by Contract Type



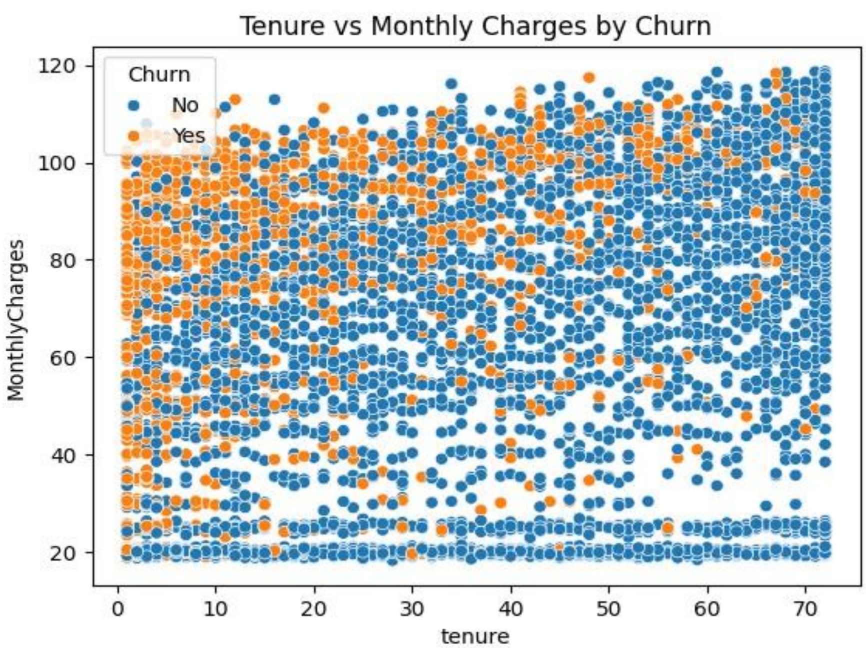
This is one of the strongest patterns in the dataset: month-to-month customers churn at a dramatically higher rate than one-year or two-year customers, whose churn counts are comparatively small. This suggests that contract length is a major driver of retention.

4.7 Correlation Heatmap



Tenure shows the strongest relationship with churn (correlation of -0.35), meaning that longer-tenured customers are notably less likely to churn. TotalCharges shows a weaker negative correlation (-0.2), while MonthlyCharges shows a weak positive correlation (0.19), indicating that customers with higher monthly bills are slightly more likely to churn.

4.8 Tenure vs Monthly Charges by Churn



Churned customers (shown in orange) are concentrated at lower tenure values across a wide range of monthly charges, reinforcing that early-tenure customers — regardless of how much they pay — are the group most at risk of leaving.

5. Key Findings

- **Overall churn rate:** 26.6% of customers churned, while 73.4% stayed, out of 7,032 cleaned customer records.
- **Highest-risk segment:** Month-to-month contract customers have by far the highest churn rate; customers on one-year and two-year contracts are much more stable.
- **Strongest numerical driver:** Tenure has the strongest correlation with churn (-0.35) — the longer a customer stays, the less likely they are to leave.
- **Pricing effect:** MonthlyCharges has a weak positive correlation with churn (0.19), suggesting higher bills are mildly associated with higher churn, but it is a much smaller effect than contract type or tenure.
- **Data quality:** The dataset required minimal cleaning — only 11 rows with missing TotalCharges values needed to be removed, and no outliers were found in any numerical column.
- **Top 3 factors driving churn:** (1) Contract type — month-to-month customers churn far more than long-term contract customers, (2) Tenure — newer customers are more likely to leave, and (3) Monthly charges — customers with higher bills show a slightly higher tendency to churn.

6. Recommendations

- **Incentivize longer contracts:** Since month-to-month customers churn the most, offer discounts, loyalty pricing, or bundled perks to encourage them to move to one-year or two-year contracts.
- **Strengthen early-tenure engagement:** Since churn risk is highest in the first few months, invest in onboarding programs, proactive check-ins, or early-tenure loyalty rewards to keep new customers engaged past the critical early period.
- **Review pricing for high-bill customers:** Since higher MonthlyCharges show a mild association with churn, consider targeted retention offers or value-added services for customers on higher-priced plans to reduce price-sensitivity-driven churn.
- **Monitor at-risk segments proactively:** Combine contract type and tenure to flag high-risk customers (e.g., month-to-month customers within their first 6 months) for proactive retention outreach.

7. Conclusion

This analysis shows that churn is not random — it is strongly concentrated among customers with month-to-month contracts and short tenure, with a smaller but still notable effect from higher monthly charges. Customers with longer contracts and longer tenure are far more likely to stay. By focusing retention efforts on contract upgrades, early-tenure engagement, and pricing strategy for high-bill customers, the business has a clear, data-backed path to reducing churn and improving long-term customer retention.